

51. TextDecoder aur TextEncoder

Agar binary data asal me **string** ho (jaise text file mili), to use JS string me padhne ke liye `TextDecoder` hai. Ulta (string se bytes) ke liye `TextEncoder`.

TextDecoder

js

```
let decoder = new TextDecoder([label], [options]);
```

- `label`: encoding. Default `utf-8`, par `big5`, `windows-1251` waghera bhi chalte hain.
- `options`:
 - `fatal`: `true` ho to invalid characters par **exception**, warna (default) unhe `\uFFFD` se badal deta hai.
 - `ignoreBOM`: byte-order mark ignore karna (kam zarurat).

Decode karna:

js

```
let str = decoder.decode([input], [options]);
```

- `input`: `BufferSource` (`ArrayBuffer` ya uska view)
- `options.stream`: `true` jab data **tukdon (chunks)** me aa raha ho. Kabhi kabhi ek multi-byte character do chunks me kat jaata hai, ye option adhoore character ko yaad rakh ke agle chunk ke saath decode karta hai.

Examples:

js

```
new TextDecoder().decode(new Uint8Array([72, 101, 108, 108, 111])); // Hello
new TextDecoder().decode(new Uint8Array([228, 189, 160, 229, 165, 189])); // 你好
```

Buffer ka sirf ek hissa decode karna ho to `subarray` (bina copy) use karo:

js

```
let uint8Array = new Uint8Array([0, 72, 101, 108, 108, 111, 0]);
let binaryString = uint8Array.subarray(1, -1);
new TextDecoder().decode(binaryString); // Hello
```

TextEncoder

String ko bytes me badalta hai. Sirf **utf-8** support karta hai.

js

```
let encoder = new TextEncoder();
encoder.encode("Hello");           // Uint8Array [72,101,108,108,111]
encoder.encodeInto(str, destination); // destination Uint8Array hona chahiye
```

Yaad rakho

- Bytes -> string: **TextDecoder**
- String -> bytes: **TextEncoder**
- Chunked data me **stream: true** zaruri.